

Week2

Problem 1

```
import java.util.Arrays;

public class Solution2_1 {

    public int[] runningSum(int[] nums) {
        for (int i = 1; i < nums.length; i++) ;
            nums[i] += nums[i - 1];
        }
        return nums;
    }

    public static void main(String[] args) {
        Solution solution = new Solution();
        int[] nums1 = {1, 2, 3, 4};
        System.out.println("Input: " + Arrays.toString(nums1));
        int[] result1 = solution.runningSum(nums1);
        System.out.println("Output: " + Arrays.toString(result1)); // Expected: [1, 3, 6, 10]
        System.out.println();

        int[] nums2 = {1, 1, 1, 1, 1};
        System.out.println("Input: " + Arrays.toString(nums2));
        int[] result2 = solution.runningSum(nums2);
        System.out.println("Output: " + Arrays.toString(result2)); // Expected: [1, 2, 3, 4, 5]
        System.out.println();

        int[] nums3 = {3, 1, 2, 10, 1};
        System.out.println("Input: " + Arrays.toString(nums3));
        int[] result3 = solution.runningSum(nums3);
        System.out.println("Output: " + Arrays.toString(result3)); // Expected: [3, 4, 6, 16, 17]
    }
}
```

Output-

Input: [1, 2, 3, 4]

Output: [1, 3, 6, 10]

Input: [1, 1, 1, 1, 1]

Output: [1, 2, 3, 4, 5]

Input: [3, 1, 2, 10, 1]

Output: [3, 4, 6, 16, 17]

Problem 2

```
import java.util.Arrays;
```

```
public class Solution2_2 {
```

```
    public int[] shuffle(int[] nums, int n) {
```

```
        int[] result = new int[2 * n];
```

```
        for (int i = 0; i < n; i++) {
```

```
            result[2 * i] = nums[i];
```

```
            result[2 * i + 1] = nums[i + n];
```

```
        }
```

```
        return result;
```

```
    }
```

```
    public static void main(String[] args) {
```

```
        Solution solution2_2 = new Solution2_2();
```

```
        int[] nums1 = {2, 5, 1, 3, 4, 7};
```

```
        int n1 = 3;
```

```
        System.out.println("Input: " + Arrays.toString(nums1) + ", n = " + n1);
```

```
        int[] result1 = solution.shuffle(nums1, n1);
```

```
        System.out.println("Output: " + Arrays.toString(result1)); // Expected: [2, 3, 5, 4, 1, 7]
```

```
        System.out.println();
```

```
        int[] nums2 = {1, 2, 3, 4, 4, 3, 2, 1};
```

```
        int n2 = 4;
```

```
        System.out.println("Input: " + Arrays.toString(nums2) + ", n = " + n2);
```

```

        int[] result2 = solution.shuffle(nums2, n2);
        System.out.println("Output: " + Arrays.toString(result2)); // Expected: [1, 4, 2, 3, 3, 2, 4,
1]
        System.out.println();
        int[] nums3 = {1, 1, 2, 2};
        int n3 = 2;
        System.out.println("Input: " + Arrays.toString(nums3) + ", n = " + n3);
        int[] result3 = solution.shuffle(nums3, n3);
        System.out.println("Output: " + Arrays.toString(result3)); // Expected: [1, 2, 1, 2]
    }
}

```

Output-

Input: [2, 5, 1, 3, 4, 7], n = 3

Output: [2, 3, 5, 4, 1, 7]

Input: [1, 2, 3, 4, 4, 3, 2, 1], n = 4

Output: [1, 4, 2, 3, 3, 2, 4, 1]

Input: [1, 1, 2, 2], n = 2

Output: [1, 2, 1, 2]

Problem 3

```

import java.util.Arrays;

public class Solution2_1 {

    public int removeElement(int[] nums, int val) {

        int k = 0;

        for (int i = 0; i < nums.length; i++) {

            if (nums[i] != val) {

                nums[k] = nums[i];

                k++;

            }

        }

    }

}

```

```

    }
    return k;
}

public static void main(String[] args) {
    Solution2_1 solution = new Solution2_1();

    // Test Case 1
    int[] nums1 = {3, 2, 2, 3};
    int val1 = 3;
    System.out.println("Input: nums = " + Arrays.toString(nums1) + ", val = " + val1);
    int k1 = solution.removeElement(nums1, val1);
    System.out.println("Output: k = " + k1);
    System.out.print("Modified Array (first k elements): [");
    for (int i = 0; i < k1; i++) {
        System.out.print(nums1[i] + (i < k1 - 1 ? ", " : ""));
    }
    System.out.println("]");
    System.out.println();

    // Test Case 2
    int[] nums2 = {0, 1, 2, 2, 3, 0, 4, 2};
    int val2 = 2;
    System.out.println("Input: nums = " + Arrays.toString(nums2) + ", val = " + val2);
    int k2 = solution.removeElement(nums2, val2);
    System.out.println("Output: k = " + k2);
    System.out.print("Modified Array (first k elements): [");
    for (int i = 0; i < k2; i++) {
        System.out.print(nums2[i] + (i < k2 - 1 ? ", " : ""));
    }
    System.out.println("]");
}
}

```

Output-

Input: nums = [3, 2, 2, 3], val = 3

Output: k = 2

Modified Array (first k elements): [2, 2]

Input: nums = [0, 1, 2, 2, 3, 0, 4, 2], val = 2

Output: k = 5

Modified Array (first k elements): [0, 1, 3, 0, 4]

Problem 4

```
import java.util.Arrays;

public class Solution2_4 {

    public int removeDuplicates(int[] nums) {
        if (nums.length == 0) {
            return 0;
        }
        int k = 0;
        for (int i = 1; i < nums.length; i++) {

            if (nums[i] != nums[k]) {
                k++;
                nums[k] = nums[i];
            }
        }
        return k + 1;
    }

    public static void main(String[] args) {
        Solution2_4 solution = new Solution2_4();

        int[] nums1 = {1, 1, 2};
```

```

System.out.println("Input: nums = " + Arrays.toString(nums1));
int k1 = solution.removeDuplicates(nums1);
System.out.println("Output: k = " + k1);
System.out.print("Modified Array (first k elements): [");
for (int i = 0; i < k1; i++) {
    System.out.print(nums1[i] + (i < k1 - 1 ? ", " : ""));
}
System.out.println("]");
System.out.println();

```

```

int[] nums2 = {0, 0, 1, 1, 1, 2, 2, 3, 3, 4};
System.out.println("Input: nums = " + Arrays.toString(nums2));
int k2 = solution.removeDuplicates(nums2);
System.out.println("Output: k = " + k2);
System.out.print("Modified Array (first k elements): [");
for (int i = 0; i < k2; i++) {
    System.out.print(nums2[i] + (i < k2 - 1 ? ", " : ""));
}
System.out.println("]");
}
}

```

Output-

Input: nums = [1, 1, 2]

Output: k = 2

Modified Array (first k elements): [1, 2]

Input: nums = [0, 0, 1, 1, 1, 2, 2, 3, 3, 4]

Output: k = 5

Modified Array (first k elements): [0, 1, 2, 3, 4]

Problem 5

```

import java.util.Arrays;

public class Solution2_5 {

    public int maxSubArray(int[] nums) {
        int maxSoFar = nums[0];
        int currentMax = nums[0];
        for (int i = 1; i < nums.length; i++) {
            currentMax = Math.max(nums[i], currentMax + nums[i]);
            maxSoFar = Math.max(maxSoFar, currentMax);
        }
        return maxSoFar;
    }

    public static void main(String[] args) {
        Solution2_5 solution = new Solution2_5();

        int[] nums1 = {-2, 1, -3, 4, -1, 2, 1, -5, 4};
        int result1 = solution.maxSubArray(nums1);
        System.out.println("Input: " + Arrays.toString(nums1));
        System.out.println("Output (Expected: 6): " + result1);
        System.out.println();

        int[] nums2 = {1};
        int result2 = solution.maxSubArray(nums2);
        System.out.println("Input: " + Arrays.toString(nums2));
        System.out.println("Output (Expected: 1): " + result2);
        System.out.println();

        int[] nums3 = {5, 4, -1, 7, 8};
        int result3 = solution.maxSubArray(nums3);
        System.out.println("Input: " + Arrays.toString(nums3));
        System.out.println("Output (Expected: 23): " + result3);
    }
}

```

```

        System.out.println();

        int[] nums4 = {-5, -1, -3, -4};
        int result4 = solution.maxSubArray(nums4);
        System.out.println("Input: " + Arrays.toString(nums4));
        System.out.println("Output (Expected: -1): " + result4);
    }
}

```

Output-

Input: [-2, 1, -3, 4, -1, 2, 1, -5, 4]

Output (Expected: 6): 6

Input: [1]

Output (Expected: 1): 1

Input: [5, 4, -1, 7, 8]

Output (Expected: 23): 23

Input: [-5, -1, -3, -4]

Output (Expected: -1): -1

Problem 6

```

import java.util.Arrays;

public class Solution2_6 {

    public int largestAltitude(int[] gain) {
        int currentAltitude = 0;
        int maxAltitude = 0;
        for (int i = 0; i < gain.length; i++) {
            currentAltitude += gain[i];
            maxAltitude = Math.max(maxAltitude, currentAltitude);
        }
    }
}

```

```

    }
    return maxAltitude;
}

public static void main(String[] args) {
    Solution2_6 solution = new Solution2_6();

    int[] gain1 = {-5, 1, 5, 0, -7};
    System.out.println("Input: gain = " + Arrays.toString(gain1));
    int result1 = solution.largestAltitude(gain1);
    System.out.println("Output: " + result1); // Expected: 1
    System.out.println();

    int[] gain2 = {-4, -3, -2, -1, 4, 3, 2};
    System.out.println("Input: gain = " + Arrays.toString(gain2));
    int result2 = solution.largestAltitude(gain2);
    System.out.println("Output: " + result2); // Expected: 0
}
}

```

Output

Input: gain = [-5, 1, 5, 0, -7]

Output: 1

Input: gain = [-4, -3, -2, -1, 4, 3, 2]

Output: 0

Problem 7

```

import java.util.*;

class Solution {
    public List<List<String>> groupAnagrams(String[] strs) {
        HashMap<String, List<String>> map = new HashMap<>();
        for (String str : strs) {
            // Convert string to character array

```

```

        char[] chars = str.toCharArray();
        // Sort characters
        Arrays.sort(chars);

        // Convert back to string
        String key = new String(chars);

        // Add string to its anagram group
        if (!map.containsKey(key)) {
            map.put(key, new ArrayList<>());
        }
        map.get(key).add(str);
    }
    return new ArrayList<>(map.values());
}
}

```

Output-

Input1

strs =

["eat","tea","tan","ate","nat","bat"]

Output

[["eat","tea","ate"],["bat"],["tan","nat"]]

Expected

[["bat"],["nat","tan"],["ate","eat","tea"]]

Input2

strs =

[""]

Output

[[""]]

Expected

[[""]]

Input3

strs =

["a"]

Output

[["a"]]

Expected

[["a"]]

Problem 8

```
import java.util.*;
```

```
class Solution {
```

```
    public int[] topKFrequent(int[] nums, int k) {
```

```
        // Count frequency of each number
```

```
        HashMap<Integer, Integer> map = new HashMap<>();
```

```
        for (int num : nums) {
```

```
            map.put(num, map.getOrDefault(num, 0) + 1);
```

```
        }
```

```
        // Bucket array
```

```
        List<Integer>[] bucket = new List[nums.length + 1];
```

```
        for (int num : map.keySet()) {
```

```
            int frequency = map.get(num);
```

```
            if (bucket[frequency] == null) {
```

```
                bucket[frequency] = new ArrayList<>();
```

```
            }
```

```
            bucket[frequency].add(num);
```

```
        }
```

```
        // Store answer
```

```
        int[] result = new int[k];
```

```
        int index = 0;
```

```
        // Start from highest frequency
```

```
        for (int i = bucket.length - 1; i >= 0 && index < k; i--) {
```

```
            if (bucket[i] != null) {
```

```
                for (int num : bucket[i]) {
```

```
        result[index] = num;
        index++;
        if (index == k) {
            break;
        }
    }
}
}
}
return result;
}
```

Output

Input 1

nums =

[1,1,1,2,2,3]

k =

2

Output

[1,2]

Expected

[1,2]

Input 2

nums =

[1]

k =

1

Output

[1]

Expected

[1]

Input 3

nums =

[1,2,1,2,1,2,3,1,3,2]

k =

2

Output

[1,2]

Expected

[1,2]